Machine Learning Assignment: Linear Regression

1 Learning objectives

- Distinguish statistical association from causation;
- Interpret regression coefficients in multivariable biological models;
- Understand R^2;
- Recognize regression assumptions and overfitting;
- Apply linear regression to datasets.

2 Problems

2.1 Problem 1: Conceptual Understanding and Model Interpretation (40 points)

A researcher studies whether the expression level of a stress-response gene is associated with oxidative stress in cultured human cells. For $n = 120$ samples, the following multiple linear regression model is fitted:

$$\begin{aligned} y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon \\ &= \beta_0 + \beta_1 \times \text{ROS} + \beta_2 \times \text{Cell age} + \beta_3 \times \text{Treatment} \end{aligned}$$

where:

- y: expression level of Gene A measured as $\log_2$-normalized expression
- x_1: intracelluar reactive oxygen (ROS) species level
- x_2: Cell age in days
- x_3: treatment status, coded 0 = control and 1 = treated.

The fitted model is:

$$\hat{y} = 2.10 + 0.035x_1 - 0.12x_2 + 0.85x_3$$

with the following results:

	Coefficient	Standard Error	p_value
Intercept	2.10	0.42	< 0.001
ROS	0.035	0.009	< 0.001
Cell age	-0.12	0.05	< 0.019
Treatment	0.85	0.21	< 0.001

The model has $R^2 = 0.48$ and adjusted $R^2 = 0.47$. Questions:

1. Interpret β_1, β_2, and β_3 in the biological context. Be precise about what is being held constant.

2. What does $R^2 = 0.48$ mean? Does it mean that the model predicts gene expression accurately for individual samples. Explain.

3. A student concludes: '*Because the coefficient for ROS is statistically significant, increasing ROS causes Gene A expression to increase.*' Explain why this conclusion is not necessarily justified.

4. Suppose a residual-versus-fitted plot shows a clear funnel shape, with residual variance increasing as fitted gene expression increases.

   - Which regression assumption is likely violated?
   - What consequences might this have?
   - Suggest one possible remedy.

5. ROS and treatment status are strongly associated because the treatment itself increases ROS. Explain how this relationship could affect interpretation of the ROS coefficient.

6. Suppose the researcher adds 50 randomly generated, biologically meaningless predictors to the model. Explain what you would generally expect to happen to:

   - R^2
   - adjusted R^2
   - training error
   - test error

2.2 Problem 2: Linear regression with a real biological dataset (60 points)

The scikit-learn Diabetes Dataset (`https://scikit-learn.org/stable/datasets/toy_dataset.html`) or Sklearn Diabetes dataset consists of ten baseline variables, such as age, sex, body mass index (BMI), average blood pressure, and six blood serum measurements, obtained for 442 diabetes patients. The target variable is a quantitative measure of disease progression one year after baseline. Figure 2.2 shows the first five rows of the dataset.

Attributes

quantitative measure of disease progression

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	target
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646	151.0
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204	75.0
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930	141.0
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362	206.0
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641	135.0

Sample Data *

* Sklearn Diabetes Dataset contains 442 rows of data

Figure 1: Sklearn Diabetes Dataset

Note: Each of these 10 feature variables have been mean centered and scaled by the standard deviation times the square root of n_samples (i.e. the sum of squares of each column totals 1).

You may load the data with:

```python
from sklearn.datasets import load_diabetes

diabetes = load_diabetes(as_frame=True)

X = diabetes.data
y = diabetes.target

print(X.shape)
print(X.head())
print(y.describe())
```

Part A. Data exploration

1. Report the number of observations and predictors.
2. Examine the distributions of the predictors and outcome.
3. Calculate correlations between each predictor and the outcome.
4. Identify the three predictors with the strongest absolute correlations with disease progression.
5. Create a scatterplot of BMI versus disease progression. Describe the apparent relationship.

Part B. Simple linear regression
Fit a model using BMI as the only predictor:

$$y = \beta_0 + \beta_1 \times \text{BMI} + \epsilon$$

1. Split the data into 80% training and 20% testing sets using random_state=42.
2. Fit the model using the training data.
3. Report the intercept and BMI coefficient.
4. Interpret the BMI coefficient. Pay attention to the fact that the predictors in this dataset have already been standardized.
5. Calculate the training and testing R^2 and RMSE.
6. Plot the observed test outcomes versus predicted outcomes.

Part C: Multiple linear regression
Fit a second model using all 10 predictors.

1. Report the estimated coefficients.
2. Compare its training and testing R^2 and RMSE with the BMI-only model.
3. Which model performs better on unseen data?
4. Identify the three predictors with the largest absolute regression coefficients. Does this necessarily mean that these variables are the three most biologically important predictors? Explain.

Part D: Model diagnostics
For the multiple regression model:

1. Plot residuals versus fitted values for the training data.
2. Plot the distribution of residuals.

3. Discuss whether the residuals appear consistent with the assumptions of constant variance and approximate normality.
4. Investigate correlations among the predictor variables. Are there predictors that may introduce multicollinearity?
5. Explain how multicollinearity can affect coefficient estimates even when overall predictive performance remains reasonable.